Communication-Efficient Federated Learning through Adaptive Weight Clustering and Server-Side Distillation

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01 Context & approach

The challenge

Compressing client models saves upload bandwidth, but averaging their weights breaks the shared-centroid structure. *FedCompress* restores compression at the server and adjusts the number of centroids as learning progresses, reducing traffic in both directions.

Our contribution

Compress client uploads and server broadcasts with weight clustering.
Adapt cluster capacity using the effective rank of local embeddings.
Measure accuracy, communication and model size across five vision and audio tasks.

1 Cluster client weights

After supervised warm-up, combine label training with centroid clustering and upload the compressed model.

2 Restore server compression

Apply FedAvg, then distill its aggregated model into a clustered student using auxiliary unlabeled data.

3 Adapt capacity

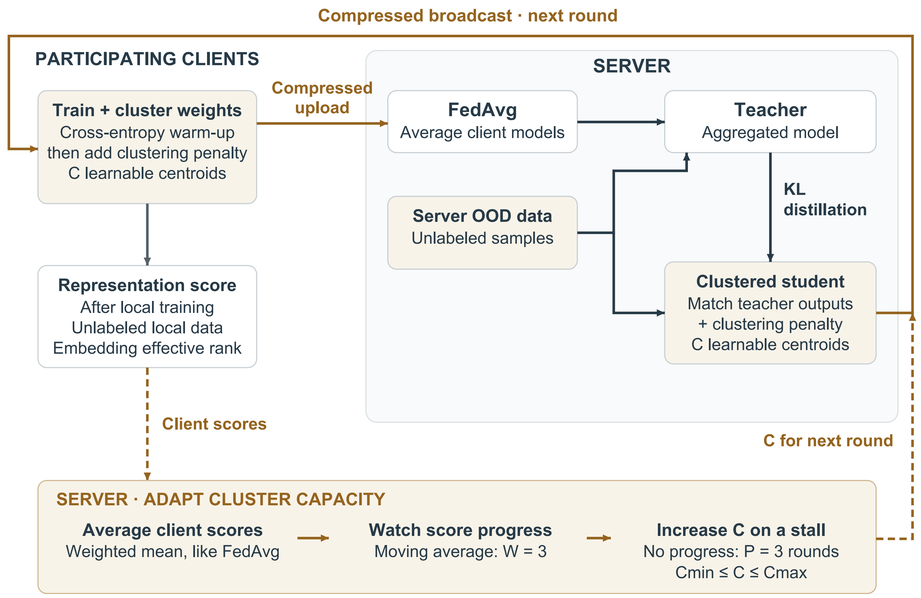
Combine clients' effective-rank scores; increase the cluster count for later rounds when score progress stalls.

Scope

Server distillation requires auxiliary data and extra optimization. Simulated clients; two-run accuracy means without reported dispersion.

02 Method

Compress both communication directions through local clustering and server distillation.



Core formulation

$$\mathcal{L}_S = \mathcal{L}_{\mathrm{KL}}(p_T \| p_\theta) + \beta \mathcal{L}_{\mathrm{wc}}(\theta, \mu, C)$$

Teacher p_T guides student p_θ; the clustering penalty groups weights around C centroids μ. β balances compression and imitation.

$$\mathcal{E}_k = \exp\left(-\sum_j r_j \log r_j\right)$$

r_j are normalized singular values of client k's embeddings. Their effective rank tracks representation diversity and guides cluster capacity.

Paper

Code

03 Results

5.41×
communication reduction
VoxForge versus FedAvg

-0.31 pp
accuracy difference
Same VoxForge comparison

Communication savings with small accuracy losses

Dataset	FedAvg (%)	Δ acc. (pp)	Comm. (×)	Model (×)
CIFAR-10	86.26	-1.83	4.53	5.18
CIFAR-100	60.68	-1.88	3.80	3.93
PathMNIST	88.22	-1.72	4.79	5.27
SpeechCommands	95.75	-0.42	5.04	5.09
VoxForge	81.05	-0.31	5.41	5.64

20 clients, 20 rounds, 10 local/server epochs; two-run mean accuracy. Signed accuracy changes and reduction ratios are relative to FedAvg.

Why compress again at the server?

CIFAR-100 method	Δ acc. (pp)	Comm. (×)	Model (×)
FedZip	-2.57	1.94	2.11
Without server compression	-2.67	1.02	1.62
FedCompress (Ours)	**-1.88**	**3.80**	**3.93**

CIFAR-100 under the same protocol; accuracy changes and reduction ratios relative to FedAvg.

Takeaway

FedCompress reduces communication across all five tasks, with **small accuracy losses**. On CIFAR-100, **server compression** raises the communication reduction from 1.02× to 3.80×.